

2.6 Differential equations

Further Pure 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Separate variables only when dy/dx can be arranged as $f(y)g(x)$. For linear first-order equations use an integrating factor when required. Particular solutions require applying the initial condition after finding the general solution; an omitted constant loses a family of solutions.

Worked example: follow the reasoning

$dy/dx = 2xy$ gives $dy/y = 2x dx$, hence $\ln|y| = x^2 + C$ and $y = Ae^{x^2}$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Solve $dy/dx = 3y$ given $y(0) = 2$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Separate $dy/y=3dx$ [1]; $\ln|y|=3x+C$ [1]; $y=2e^{(3x)}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.